MATHEMATICS SINGLE CORRECT | MATHEMATICS NUMERICAL | PHYSICS SINGLE CORRECT | PHYSICS NUMERICAL | CHEMISTRY SINGLE C

Question No. 1 #6776c0da4a3f794b86e91b991

Section Details | Marking Scheme

Mathematics Single Correct (Maximum Marks 80)

Only One Option Correct Type

This section contains 20 questions. 20 are multiple choice questions. Each question has multiple options out of which ONLY ONE is correct.

If T_m&T_n denotes the mth & n^{th} terms of an A.P. respectively, such that $T_m = \frac{1}{n}$, $T_n = \frac{1}{m}$, then which of the following is necessarily root of the equation $(a-3b)x^2 + (2b+5a)x + (b-6a) = 0$:

- T_{mn}
- T_{m+n}
- $T_m + T_n$
- $T_m \cdot T_n$

All Sections | Time Left: 178:24

MATHEMATICS SINGLE CORRECT | MATHEMATICS NUMERICAL | PHYSICS SINGLE CORRECT | PHYSICS NUMERICAL | CHEMISTRY SINGLE CO

Question No. 2 #6776c0da4a3f794b86e91b992 ... Marking Scheme >

Mathematics Single Correct (Maximum Marks: 80)
Only One Option Correct Type

This section contains 20 questions. 20 are multiple choice questions. Each question has multiple options out of which ONLY ONE is correct.

The solution of the differential equation $\frac{dy}{dx} + x(2x+y) = x^3(2x+y)^3 - 2$ is (C being an arbitrary constant)

- $\frac{1}{2x+xy} = x^2 + 1 + Ce^x$
- $\frac{1}{(2x+y)^2} = x^2 + 1 + Ce^x$
- $\frac{1}{2x+y} = x + 1 + Ce^{-x^2}$

MATHEMATICS SINGLE CORRECT | MATHEMATICS NUMERICAL | PHYSICS SINGLE CORRECT | PHYSICS NUMERICAL | CHEMISTRY SINGLE COR

Question No. 3 #6776c0da4a3f794b86e91b993

Section Details | Marking Scheme

Mathematics Single Correct (Maximum Marks: 80)
Only One Option Correct Type

This section contains 20 questions. 20 are multiple choice questions. Each question has multiple options out of which ONLY ONE is correct.

Suppose we have 7 coins (C_i), where $i = 1, 2, \ldots\ldots 7$ such that the probability of head on i^{th} coin is $P(H_i) = \frac{i+1}{8}$. One of the coin is randomly selected and tossed and H shows the event that it shows up the head. Then which of the following is true -

- $\mathrm{P(C_4/H)} = \frac{1}{7}$
- $\mathrm{P(C_1/H)} = \frac{3}{35}$
- $\mathrm{P(H)} = \frac{1}{4}$
- $\mathrm{P(\overline{H})} = \frac{5}{8}$

Mark for Review & Next

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MATHEMATICS SINGLE CORRECT | MATHEMATICS NUMERICAL | PHYSICS SINGLE CORRECT | PHYSICS NUMERICAL | CHEMISTRY SINGLE COR

Question No. 4 #6776c0da4a3f794b86e91b994

Section Details | Marking Scheme

Mathematics Single Correct (Maximum Marks: 80)
Only One Option Correct Type

This section contains 20 ...estions. 20 are multiple ... ...ce questions. Each question has multiple options out of which ONLY ONE is correct.

Let α, β and γ be three distinct real roots of the equation $x(3x+2)^2+2=(a+12+9x)x^2-bx+c$ where $a, b, c \in R$. If every solution of the inequality $(x-a)^2(4x+b)(x-c)<0$ is also solution of the inequality $3x^2+px+p^2+6p<0$ then find number of integral values of 'p'-

- 3
- 4
- 5
- 6

Mark for Review & Next ...espons

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MATHEMATICS SINGLE CORRECT | MATHEMATICS NUMERICAL | PHYSICS SINGLE CORRECT | PHYSICS NUMERICAL | CHEMISTRY SINGLE CO

Question No. 5 #6776c0da4a3f794b86e91b995

Section Details | Marking Scheme

Mathematics Single Correct (Maximum Marks: 80)
Only One Option Correct Type

.s section contains 20 questions. 20 are multiple choice questions. Each q .ultiple options out of which ONLY ONE is correct.

Let l_1 be the line $4x + 3y = 3$ and I_2 be the line $y = 8x$. L_1 is the line formed by reflecting l_1 across the line $y = x$ and L_2 is the line formed by reflecting I_2 across the x-axis. If θ is the acute angle between L_1 and L_2 such that $\tan\theta = \frac{a}{b}$, where a and b are coprime then find $(a + b)$?

- 71
- 63
- 57
- 27

Mark for Review & Next | esponse

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MATHEMATICS SINGLE CORRECT | MATHEMATICS NUMERICAL | PHYSICS SINGLE CORRECT | PHYSICS NUMERICAL | CHEMISTRY SINGLE CO

Question No. 6 #6776c0da4a3f794b86e91b996

Section Details | Marking Scheme

Mathematics Single Correct (Maximum Marks: 80)
Only One Option Correct Type

This section contains 20 questions. 20 are multiple choice questions. Each question nas multiple options out of which ONLY ONE is correct.

Let $f\left(x\right)=\frac{x}{\left(1+x^n\right)^{\frac{1}{n}}}, n\geq 2,\ g\left(x\right)=\left(\underbrace{fofo\ldots.\text{ of}}_{n\ times}\right)\left(x\right)$ and $I\left(n,\ x\right)=\int g\left(x\right)\cdot x^{n-2}dx$, If $I\left(10,1\right)=\frac{\left(p\right)^{\frac{m}{k}}}{q}$, then the value of $10\ k-m-q+p=$_____ (Given that G.C.D. $\left(m,\ k\right)=1$))

- 11
- 10
- 8
- 12

Mark for Review & Next

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MATHEMATICS SINGLE CORRECT | MATHEMATICS NUMERICAL | PHYSICS SINGLE CORRECT | PHYSICS NUMERICAL | CHEMISTRY SINGLE CO

Question No. 7 #6776c0da4a3f794b86e91b997

Section Details | Marking Scheme

Mathematics Single Correct (Maximum M)
Only One Option Correct T

...is section contains 20 questions. 20 are multiple choice questions. Each ques ...s multiple options out ... ...n ONLY ONE is correct.

Let S be the focus of the parabola $y^2 = 8x$ and PQ be the common chord of the circle $x^2 + y^2 - 2x - 4y = 0$ and the given parabola. The area of ΔPQS is.

- 4sq units
- 3sq units
- 2 sq units
- $8sq$ units

Mark for Review & Next

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MATHEMATICS SINGLE CORRECT | MATHEMATICS NUMERICAL | PHYSICS SINGLE CORRECT | PHYSICS NUMERICAL | CHEMISTRY SINGLE COR

Question No. 8 #6776c0da4a3f794b86e91b998

Section Details | Marking Scheme

Mathematics Single Correct (Maximum Marks: 80)
Only One Option Correct Type

nis section contains 20 questions. 20 are multiple choice questions. Each question has multiple options ou ich ONLY ONE is correct.

Let $L_1 : \frac{x-6}{3} = \frac{y-7}{-1} = \frac{z-4}{1}$ and $L_2 : \frac{x}{-3} = \frac{y+9}{2} = \frac{z-2}{4}$. If $L_1 \& L_2$ are skew lines such that $P \& Q$ are two points nearest to each other lying on lines $L_1 \& L_2$ respectively. If coordinate of P is (a,b,c), find $a + 2b + 3c$.

- 28
- 36
- 24
- 32

Mark for Review & Next

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MATHEMATICS SINGLE CORRECT | MATHEMATICS NUMERICAL | PHYSICS SINGLE CORRECT | PHYSICS NUMERICAL | CHEMISTRY SINGLE COR

Question No. 9 #6776c0da4a3f794b86e91b999

Section Details | Marking Scheme

Mathematics Single Correct (Maximum Mark 80)
Only One Option Correct Type

This section contains 20 questions. 20 are multiple choice questions. Each question has multiple options out which ONLY ONE is correct.

Let $\mathbf{f}:[-1,2] \rightarrow [0,\infty)$ be a continuous function such that $\mathrm{f}(x)=\mathrm{f}(1-x), \forall x \in [-1,2]$

Let $\mathrm{R}_1=\int_{-1}^{2} x\mathrm{f}(x)\mathrm{d}x$ and R_2 be the area of the region bounded by $y=\mathrm{f}(x), x=-1, x=2$ and the X-axis, then R_2 is____

- $\frac{1}{2} R_1$
- $2R_1$
- $3R_1$
- $\frac{1}{3} R_1$

Mark for Review & Next

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MATHEMATICS SINGLE CORRECT | MATHEMATICS NUMERICAL | PHYSICS SINGLE CORRECT | PHYSICS NUMERICAL | CHEMISTRY SINGLE COR

Question No. 10 #6776c0da4a3f794b86e91b9910

Section Details | Marking Scheme

Mathematics Single Correct (Maximum Marks: 80)
Only one Option Correct Type

This section contains 20 questions. 20 are multiple choice questions. Each question has multiple options out of which ONLY ONE is correct.

The number of words which can be formed using all the letters of the word "NARENDRABHAI" such that no two non-repeated letters occur together is:

- $49(5!)$
- $48(5!)^2$
- $98(5!)$
- $98(5!)^2$

Mark for Review & Next

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MATHEMATICS SINGLE CORRECT | MATHEMATICS NUMERICAL | PHYSICS SINGLE CORRECT | PHYSICS NUMERICAL | CHEMISTRY SINGLE COR

Question No. 11 #6776c0da4a3f794b86e91b9911

Section Details | Marking Scheme

Mathematics Single Correct (Maximum Mar 0)
O Option Correct Type

his section contains 20 questions. 20 are multipl questions. Each que as multiple options ou ch ONLY ONE is correct.

The solution set of inequality
$(\tan^{-1}x)(\cot^{-1}x) - (\tan^{-1}x)\left(1+\frac{\pi}{2}\right) - 2\cot^{-1}x + 2\left(1+\frac{\pi}{2}\right) > \lim_{x\to-\infty}\left[\sec^{-1}x - \frac{\pi}{2}\right]$ is (where [.] denotes the greatest integer function)....

$(\tan 1, \tan 2)$

$(-\cot 1, \cot 2)$

○ $(-\tan 1, \tan 2)$

○ $(-\tan 1, \infty)$

Mark for Review & Next

e

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Mathematics Single Correct (Maximum Marks: 80)

This section contains 20 questions. Each question has multiple options out of which ONLY ONE is correct.

The area of the region included between the curves $x^2 + y^2 = a^2$ and $\sqrt{|x|} + \sqrt{|y|} = \sqrt{a}(a > 0)$ is---

- $\left(\pi - \frac{2}{3}\right)a^2$ sq units
- $\left(\frac{2}{3} - \pi\right)a^2$ sq units

Question No. 13 #6776c0da4a3f794b86e91b9913

Section Details | Marking Scheme

Mathematics Single Correct (Maximum Mark... 4)
One One Option Correct Type

This section contains 20 questions. 20 are multiple choice questions. Each question has multiple options out... ONLY ONE is correct.

If e_1 is the eccentricity of the ellipse $\frac{x^2}{16} + \frac{y^2}{25} = 1$ and e_2 is the eccentricity of the hyperbola passing through the foci of the ellipse and $e_1 e_2 = 1$, then the equation of the hyperbola is.

- $\frac{x^2}{9} - \frac{y^2}{16} = 1$
- $\frac{x^2}{16} - \frac{y^2}{9} = -1$
- $\frac{x^2}{9} - \frac{y^2}{25} = 1$
- None of these

Mark for Review & Next | Clear Response

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MATHEMATICS SINGLE CORRECT | MATHEMATICS NUMERICAL | PHYSICS SINGLE CORRECT | PHYSICS NUMERICAL | CHEMISTRY SINGLE COR

Question No. 14 #6776c0da4a3f794b86e91b9914

Section Details | Marking Scheme

Mathematics Sinect Correct (Maximum Marks: 30)
Only one Option Correct Type

This section contains 2 questions. 20 are multiple questiquestions. Each question has multiple options out of which ONLY ONE is correct.

If $\sum_{r=1}^{n} t_r = \frac{n(n+1)(n+2)(n+3)}{8}$, where t_r denotes the rth term of a series, then $\lim_{n\to\infty} \sum_{r=1}^{n} \frac{1}{t_r}$ is---

- $\frac{1}{8}$
- $\frac{1}{4}$
- $\frac{1}{2}$
- 1

Mark for Review & Next | Clear Response | 00:01:48 | Save & Next

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MATHEMATICS SINGLE CORRECT | MATHEMATICS NUMERICAL | PHYSICS SINGLE CORRECT | PHYSICS NUMERICAL | CHEMISTRY SINGLE COR

Question No. 15 #6776c0da4a3f794b86e91b9915

Section Details | Marking Scheme >

Mathematics Single Correct (Maximum Marks: 80)
Only One Option Correct Type

This section contains 25 questions. 20 are multiple choice questions. Each question has multiple options out of which ONLY ONE is correct.

A player tosses 2 fair coins. He wins ₹5 if 2 heads appear, ₹ 2 if one head appears and ₹ 1 if no head appears. Then the variance of his winning amount in ₹ is:

- $\frac{6}{5}$
- $\frac{5}{2}$
- $\frac{9}{4}$
- $\frac{17}{2}$

Mark for Review & Next | Clear Response

00:01:52

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MATHEMATICS SINGLE CORRECT | MATHEMATICS NUMERICAL | PHYSICS SINGLE CORRECT | PHYSICS NUMERICAL | CHEMISTRY SINGLE COR

Question No. 16 #6776c0da4a3f794b86e91b9916

Section Details | Marking Scheme

Mathematics Single Correct (Maximum Marks: 80)
Section One Option Correct Type

This section contains 20 questions. 20 are multiple choice questions. Each question has multiple options out of which ONLY ONE is correct.

Let $A = \{2, 3, 4, 5, \ldots, 30\}$ and ' $\simeq$ ' be an equivalence relation on $A \times A$, defined by $(a, b) \simeq (c, d)$, if and only if $ad = bc$. Then the number of ordered pairs which satisfy this equivalence relation with ordered pair $(4, 3)$ is equal to :

- 5
- 6
- 8
- 7

Mark for Review & Next | Clear Response

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Mathematics Single Correct (Maximum Marks: 80)

This Section Contains 20 Multiple Option Correct Type questions

This section contains 20 questions. 20 are multiple option correct questions. Each question has multiple options out of which ONLY ONE is correct.

Q

Let triangle ABC is right-angled at A. The circle with centre A and radius AB cuts BC and AC internally at D and E respectively. If $BD = 20$ units and $DC = 16$ units, then the length AB is equal to

- $6\sqrt{21}$ units
- $6\sqrt{10}$ units
- $\sqrt{300}$ units
- 18 units

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MATHEMATICS SINGLE CORRECT | MATHEMATICS NUMERICAL | PHYSICS SINGLE CORRECT | PHYSICS NUMERICAL | CHEMISTRY SINGLE COR

Question No. 18 #6776c0da4a3f794b86e91b9918

Section Details | Marking Scheme

Mathematics Single Correct (Maximum Marks: 80)
Only One Option Correct Type

This section contains 20 questions. 20 are multiple choice questions. Each question has multiple options out of which ONLY ONE is correct.

If $\alpha,\ \beta,\ \gamma$ are the number of elements in sets A, B and C respectively, and sum of number of all subsets of A, B and C is 28, evaluate the maximum number of ordered triplets $(\alpha,\ \beta,\ \gamma)$ that can be formed.

- ○ 1
- ○ 2
- ○ 4
- ○ 6

Mark for Review & Next | Clear Response

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MATHEMATICS SINGLE CORRECT | MATHEMATICS NUMERICAL | PHYSICS SINGLE CORRECT | PHYSICS NUMERICAL | CHEMISTRY SINGLE COR

Question No. 19 #6776c0da4a3f794b86e91b9919

Section Details | Marking Scheme

Mathematics Single Correct (Maximum Marks: 80)
Only One Option Correct Type

This section contains 20 questions. 20 are multiple choice questions. Each question has multiple options out of which ONLY ONE is correct.

The area (in sq. units) enclosed by the solution set of $[x][y] = 3$ is equal to (where $[\cdot]$ denotes greatest integer function)

- 5
- 3
- 4
- 2

Mark for Review & Next | Clear Response | 00:02:08 | [6] | Save & Next

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MATHEMATICS SINGLE CORRECT | MATHEMATICS NUMERICAL | PHYSICS SINGLE CORRECT | PHYSICS NUMERICAL | CHEMISTRY SINGLE COR

Question No. 20 #6776c0da4a3f794b86e91b9920

Section Details | Marking Scheme

Mathematics Single Correct (Maximum Marks: 80)
Only One Option Correct Type

This section contains 20 questions. 20 are multiple choice questions. Each question has multiple options out of which ONLY ONE is correct.

If complex numbers z_1 and z_2 both satisfy $z + \bar{z} = 2|z - 1|$ and $\arg(z_1 - z_2) = \frac{\pi}{3}$, then value of $\mathrm{Im}(z_1 + z_2)$ is, where $\mathrm{Im}(z)$ denotes imaginary part of z is $K/\sqrt{3}$, find K.

- 6
- 4
- 3
- 2

Mark for Review & Next | Clear Response | Save & Next

movavi Screen Recorder

Mathematics Numerical (Maximum Marks: 20)
Numerical Type

This section contains 5 questions. 5 are numerical questions. The answer has to be filled into the input box provided below.

Let M be a square matrix of order 3 whose elements are real numbers and $adj(adjM) = \begin{bmatrix} 36 & 0 & -4 \\ 0 & 6 & 0 \\ 0 & 3 & 6 \end{bmatrix}$, then the absolute value of $Tr(M)$ is

[Here, $adjP$ denotes adjoint matrix of P and $T_r(P)$ denotes trace of matrix P i.e. sum of all principal diagonal elements of matrix P]

Backspace

7 8 9

4 5 6

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MATHEMATICS SINGLE CORRECT | **MATHEMATICS NUMERICAL** | PHYSICS SINGLE CORRECT | PHYSICS NUMERICAL | CHEMISTRY SINGLE COR

Question No. 2 #6776c0da4a3f794b86e91b992

Section Details | Marking Scheme

Mathematics Numerical (Maximum Marks: 20)
Numerical Type

This section contains 5 questions. 5 are numerical questions. The answer has to be filled into the input box provided below.

If $(1+x)^n = C_0 + C_1x + C_2x^2 + \ldots + C_nx^n$ and $\sum_{r=0}^{50} \frac{C_r^2}{(r+1)} = \frac{m!}{(n!)^2}$, then the value of $(m+n)$ is equal to (where C_r represents C_r^n)

Backspace

7 8 9
4 5 6
1 2 3
0 . -

Mark for Review & Next | Clear Response | Save & Next

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MATHEMATICS SINGLE CORRECT | **MATHEMATICS NUMERICAL** | PHYSICS SINGLE CORRECT | PHYSICS NUMERICAL | CHEMISTRY SINGLE COR

Question No. 3 #6776c0da4a3f794b86e91b993

Section Details | Marking Scheme

Mathematics Numerical (Maximum Marks: 20)
Numerical Type

This section contains 5 questions. 5 are numerical questions. The answer has to be filled into the input box provided below.

If $\vec{a}$ and $\vec{b}$ are vectors such that $|\vec{a}| = 2, |\vec{b}| = 1, \vec{a} \wedge \vec{b} = \frac{\pi}{3}$ and $\vec{c}$ satisfies $\vec{a} + 3\vec{b} + \vec{c} = \vec{a} \times \vec{b}$, then the value of $|\vec{a} \times \vec{c}|^2$ is -

Backspace

7 8 9

4 5 6

1 2 3

0 . -

Mark for Review & Next | Clear Response | Save & Next

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Question No. 4 #6776c0da4a3f794b86e91b994

Section Details Marking Scheme

Mathematics Numerical (Maximum Marks: 20)
Numerical Type

This section contains 5 questions. 5 are numerical questions. The answer has to be filled into the input box provided below.

Consider, $f(x) = \frac{(x^2-1)(x^2-4)}{g(x)}$, $x \in R - \{a, b\}$, where g(x) is a polynomial of degree ≤ 4.

x = a, b are not in the domain of $f(x)$ but limit at these point exist.

The following three informations are also known

(i) $\text{Lim}_{x \to \pm\infty} f(x) = +\infty$

(ii) $\text{Lim}_{x \to -1} f(x) = 6$

(iii) $\text{Lim}_{x \to -2} f(x) =$ a non-zero finite number

Let L and M denotes the number of points of discontinuity and the number of points of non-differentiability of $y = |f(x)|$ respectively in R (the set of all real numbers). the value of (L + M) is

Mark for Review & Next | Clear Response | 00:02:33 | Save & Next

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Question No. 5 #6776c0da4a3f794b86e91b995

Section Details Marking Scheme

Mathematics Numerical (Maximum Marks: 20)
Numerical Type

This section contains 5 questions. 5 are numerical questions. The answer has to be filled into the input box provided below.

If the lines $\frac{x}{1}=\frac{y-1}{-1}=\frac{z+2}{2}$, $\frac{x}{2}=\frac{y-1}{1}=\frac{z+2}{1}$, $\frac{x}{-1}=\frac{y-1}{-1}=\frac{z+2}{1}$ intersect $x+y-z=0$ at P, Q, R respectively, then the area (in sq. units) of triangle PQR is $\left(\frac{9}{8}\right)\sqrt{K}$, Find K.

Backspace

7 8 9

4 5 6

1 2 3

0 . -

Left Right

Clear All

Mark for Review & Next

Clear Response

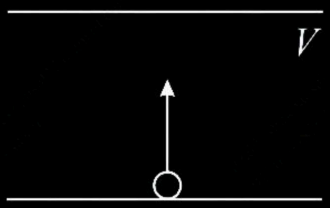
00:02:36

Save & Next

movavi Screen Recorder

This section contains 20 questions. 20 are multiple choice questions. Each question has multiple options out of which ONLY ONE is correct.

An electron having charge e and mass m starts from the lower plate of two metallic plates separated by a distance d. If the potential difference between the plates is V, the time taken by the electron to reach the upper plate is given by



- $\sqrt{\frac{2md^2}{eV}}$
- $\sqrt{\frac{md^2}{eV}}$
- $\sqrt{\frac{md^2}{2eV}}$
- $\frac{2md^2}{eV}$

Mark for Review & Next | Clear Response

movavi Screen Recorder

Question No. 2 #6776c0da4a3f794b86e91b992

Section Details Marking Scheme

Physics Single Correct (Maximum Marks: 80)
Only One Option Correct Type

This section contains 20 questions. 20 are multiple choice questions. Each question has multiple options out of which ONLY ONE is correct.

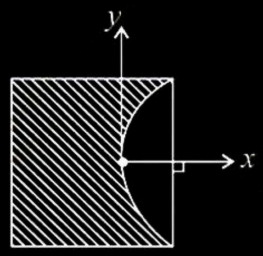
Assertion (A) When a wire of aluminium and another wire of silicon are heated from room temperature to $80^{\circ}C$, the conductivity of aluminium increases and that of silicon decreases.
Reason (R) Aluminium has positive temperature coefficient of resistivity and silicon has negative temperature coefficient of resistivity.

- Both (A) and (R) are correct and (R) is the correct explanation of (A).
- Both (A) and (R) are correct but (R) is not the correct explanation of (A).
- (A) is correct but (R) is not correct.
- (A) is not correct but (R) is correct.

Mark for Review & Next | Clear Response | Save & Next

movavi Screen Recorder

In the figure shown a semicircular area is removed from a uniform square plate of side ℓ and mass (before removing) m. The x −coordinate of centre of mass of remaining portion is (The origin is at the centre of square)



- $-\frac{\pi(\pi-2)\ell}{2(8-\pi)}$
- $\frac{\pi(\pi-2)\ell}{2(8-\pi)}$
- $-\frac{\pi(\pi-2)\ell}{(8-\pi)}$
- None of these

Mark for Review & Next | Clear Response | Save & Next

movavi Screen Recorder

Question No. 4 #6776c0da4a3f794b86e91b994

Section Details Marking Scheme

Physics Single Correct (Maximum Marks: 80)
Only One Option Correct Type

This section contains 20 questions. 20 are multiple choice questions. Each question has multiple options out of which ONLY ONE is correct.

1 kg of ice at -20°C is mixed with 2 kg of water at 90°C. Assuming that there is no loss of energy to the environment, what will be the final temperature of the mixture? (Assume latent heat of ice $= 334.4\ \text{kJ kg}^{-1}$, specific heat of water and ice are $4.18\ \text{kJ kg}^{-1}\ \text{K}^{-1}$ and $2.09\ \text{kJ kg}^{-1}\ \text{K}^{-1}$, respectively)

- 30°C
- 0°C
- 80°C
- 45°C

Mark for Review & Next | Clear Response | 00:03:00 | Save & Next

movavi Screen Recorder

Question No. 5 #6776c0da4a3f794b86e91b995

Section Details Marking Scheme

Physics Single Correct (Maximum Marks: 80)
Only One Option Correct Type

This section contains 20 questions. 20 are multiple choice questions. Each question has multiple options out of which ONLY ONE is correct.

The circumference of the second orbit of an atom or ion having single electron, is 4×10^{-9} m. The de Broglie wavelength of electron revolving in this orbit should be:

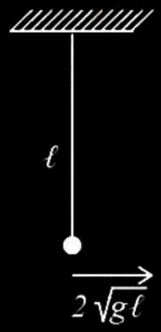
- 2×10^{-9} m
- 4×10^{-9} m
- 8×10^{-9} m
- 1×10^{-9} m

Mark for Review & Next | Clear Response | Save & Next

movavi Screen Recorder

This section contains 25 questions. 25 are multiple choice questions. Each question has multiple options out of which ONLY ONE is correct.

One end of a string of length $\ell = \frac{14}{9}$ m is fixed and a mass of 1 kg is tied to the other end. The ball is given a velocity $2\sqrt{g\ell}$ at the bottom most point as shown in figure. The string is cut when the ball becomes horizontal. Find the distance (in m) travelled till it stop for the 1st time $\left(\text{Take}\pi = \frac{22}{7}\right)$.



- 3 m
- 2 m
- 5 m
- 4 m

Mark for Review & Next | Clear Response | Save & Next

movavi Screen Recorder

Question No. 7 #6776c0da4a3f794b86e91b997

Section Details Marking Scheme

Physics Single Correct (Maximum Marks: 80)
Only One Option Correct Type

This section contains 20 questions. 20 are multiple choice questions. Each question has multiple options out of which ONLY ONE is correct.

A professor reads a greeting card received on his 50th birthday with + 2.5 D glasses keeping the card 25 cm away. Ten years later, he reads his farewell letter with the same glasses but he has to keep the letter 50 cm away. What power of lens should he now use?

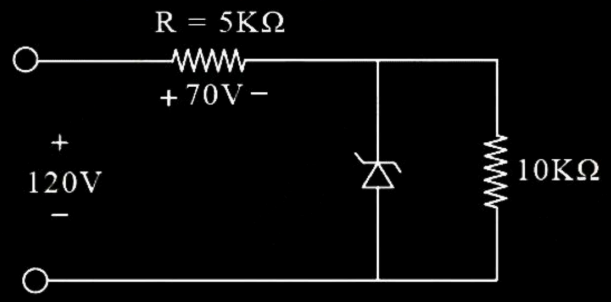
- 3D
- 3.5D
- 4D
- 4.5D

Mark for Review & Next | Clear Response

movavi Screen Recorder

This section contains 20 questions. 20 are multiple choice questions. Each question has multiple options out of which ONLY ONE is correct.

For the circuit shown in the figure :

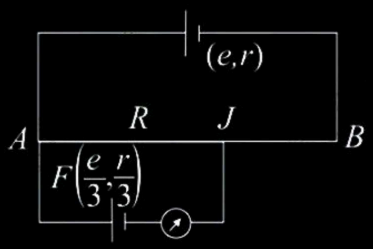


- Current through Zener diode is 4 mA
- Current through Zener diode is 9 mA
- The output voltage is 60 V
- The output voltage is 40 V

Mark for Review & Next | Clear Response | Save & Next

movavi Screen Recorder

A potentiometer arrangement is shown in the figure. The driver cell has emf e and internal resistance r.



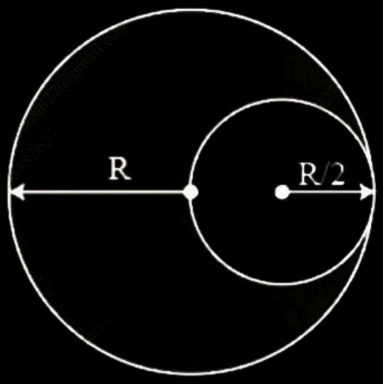
The resistance of potentiometer wire AB is R. F is the cell of emf $\frac{e}{3}$ and internal resistance $\frac{r}{3}$. Balance point (J) can be obtained for all finite values of

- $R > r/2$
- $R < r/2$
- $R > r/3$
- $R < r/3$

Mark for Review & Next | Clear Response

movavi Screen Recorder

A spherical hole is made in a solid sphere of radius R. The mass of the sphere before hollowing was M. The gravitational field at the centre of the hole due to the remaining mass is



- Zero
- $\frac{GM}{8R^2}$
- $\frac{GM}{2R^2}$
- $\frac{GM}{R^2}$

Mark for Review & Next | Clear Response | Save & Next

movavi Screen Recorder

Question No. 11 #6776c0da4a3f794b86e91b9911

Section Details Marking Scheme

Physics Single Correct (Maximum Marks: 80)
Only One Option Correct Type

This section contains 20 questions. 20 are multiple choice questions. Each question has multiple options out of which ONLY ONE is correct.

A closed organ pipe of length ' L_c ' and an open organ pipe of length ' L_0 ' contain different gases of densities ' ρ_1 ' and ' ρ_2 ' respectively. The compressibility of the gases is the same in both the pipes. The gases are vibrating in their first overtone with the same frequency. What is the length of open organ pipe?

- $\frac{4L_c}{3}\sqrt{\frac{\rho_1}{\rho_2}}$
- $\frac{3L_c}{4}\sqrt{\frac{\rho_2}{\rho_1}}$
- $\frac{4L_c}{3}\sqrt{\frac{\rho_2}{\rho_1}}$
- $\frac{2L_c}{3}\sqrt{\frac{\rho_2}{\rho_1}}$

Mark for Review & Next | Clear Response | Save & Next

movavi Screen Recorder

Question No. 12 #6776c0da4a3f794b86e91b9912

Section Details Marking Scheme

Physics Single Correct (Maximum Marks: 80)
Only One Option Correct Type

This section contains 20 questions. 20 are multiple choice questions. Each question has multiple options out of which ONLY ONE is correct.

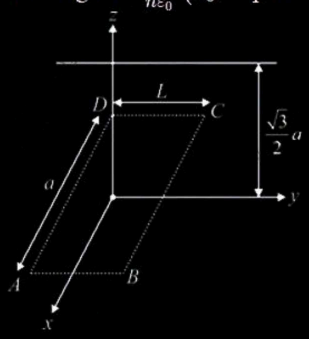
Electric displacement is given by $D = \varepsilon E$, here, ε =electric permittivity, E =electric field strength
The dimensions of electric displacement are

- $\left[\mathrm{ML}^{-2}\mathrm{TA}\right]$
- $\left[\mathrm{L}^{-2}\mathrm{T}^{-1}\mathrm{A}\right]$
- $\left[\mathrm{L}^{-2}\mathrm{TA}\right]$
- None of these

Mark for Review & Next | Clear Response | Save & Next

movavi Screen Recorder

An infinitely long uniform line charge distribution of charge per unit length λlies parallel to the y-axis in the $y-z$ plane at $z=\frac{\sqrt{3}}{2}a$ (see figure). If the magnitude of the flux of the electric field through the rectangular surface A B C D lying in the $x-y$ plane with its center at the origin is $\frac{\lambda L}{n\varepsilon_0}$ (ε_0 = permittivity of free space), then the value of n is



- 3
- 4
- 6
- 8

Mark for Review & Next | Clear Response | Save & Next

movavi Screen Recorder

Question No. 14 #6776c0da4a3f794b86e91b9914

Section Details Marking Scheme

Physics Single Correct (Maximum Marks: 80)
Only One Option Correct Type

This section contains 20 questions. 20 are multiple choice questions. Each question has multiple options out of which ONLY ONE is correct.

In a slide caliper, $(m+1)$ number of vernier divisions is equal to m number of smallest main scale divisions. If d unit is the magnitude of the smallest main scale division, then the magnitude of the vernier constant is

- $d/(m+1)$ unit
- d/m unit
- md/ $(m+1)$ unit
- $(m+1)d/m$ unit

Mark for Review & Next | Clear Response | Save & Next

movavi Screen Recorder

Question No. 15 #6776c0da4a3f794b86e91b9915

Section Details | Marking Scheme

Physics Single Correct (Maximum Marks: 80)
Only One Option Correct Type

This section contains 20 questions. 20 are multiple choice questions. Each question has multiple options out of which ONLY ONE is correct.

Assertion- If light continuously falls on a metal surface, then the emission of electrons will stop after some time.

Reason- We cannot extract all the electrons of a metal.

- Both Assertion and Reason are true and Reason is the correct explanation of Assertion.
- Assertion is true but Reason is false.
- Assertion is false but Reason is true.
- Both Assertion and Reason are false.

Mark for Review & Next | Clear Response | Save & Next

movavi Screen Recorder

Question No. 16 #6776c0da4a3f794b86e91b9916

Section Details Marking Scheme

Physics Single Correct (Maximum Marks: 80)
Only One Option Correct Type

This section contains 20 questions. 20 are multiple choice questions. Each question has multiple options out of which ONLY ONE is correct.

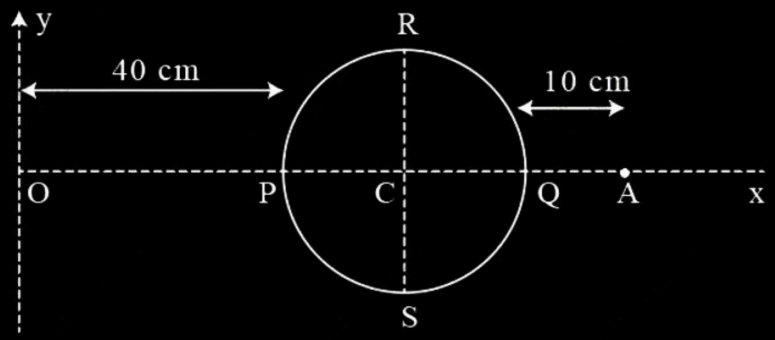
Two spherical stars A and B emit blackbody radiation. The radius of A is 400 times that of B and A emits 10^4 times the power emitted from B. The ratio (λ_A/λ_B) of their wavelengths λ_A and λ_B at which the peaks occur in their respective radiation curves is

- $\frac{1}{2}$
- $\frac{1}{4}$
- 4
- 2

Mark for Review & Next | Clear Response | Save & Next

movavi Screen Recorder

The image of a point source, kept at origin O, forms at point A due to transparent sphere of radius 10 cm as shown in the figure. Here OX is the optical axis of the sphere. Now, the sphere is cut into two equal parts, along a plane perpendicular to the horizontal axis OX and the plane surface of the left half (PRCS) is silvered. The right half (QSCR) is removed. Now the co-ordinate of image of point source O formed by remaining silvered hemisphere will be (Given $OP = 40$ cm and $QA = 10$ cm)



- $(10\text{ cm}, 0)$
- $(20\text{ cm}, 0)$
- $(30\text{ cm}, 0)$
- $(40\text{ cm}, 0)$

Mark for Review & Next | Clear Response

movavi Screen Recorder

Physics Single Correct (Maximum Marks: 80)
Only One Option Correct Type

This section contains 20 questions. 20 are multiple choice questions. Each question has multiple options out of which ONLY ONE is correct.

Assertion: In Young's double slit experiment, we observe an interference pattern on the screen if both the slits are illuminated by two bulbs of same power.

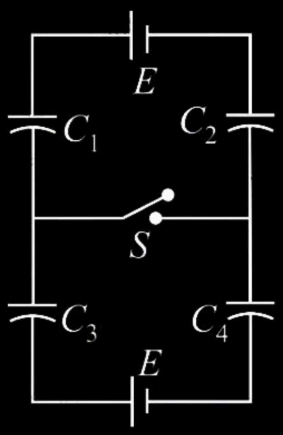
Reason: The interference pattern is observed when source is monochromatic and coherent.

- If both Assertion and Reason are true and the Reason is correct explanation of the Assertion.
- If both Assertion and Reason are true but Reason is not explanation of the Assertion.
- If Assertion is true but the Reason is false.
- If Assertion is false but Reason is true.

Mark for Review & Next | Clear Response | Save & Next

movavi Screen Recorder

In the given circuit diagram, $E = 12$ V, $C_1 = 4\ \mu$F, $C_2 = 2\ \mu$F, $C_3 = 6\ \mu$F and $C_4 = 3\ \mu$F. Find the heat produced in the circuit (in μJ) after switch S is shorted.

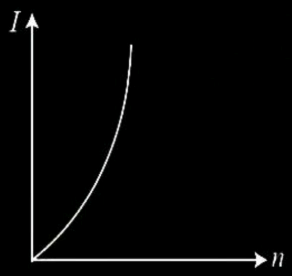


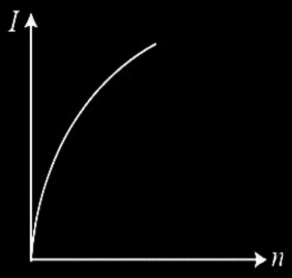
- 120 μJ
- 180 μJ
- 240 μJ
- 300 μJ

Mark for Review & Next | Clear Response | 00:04:05 | Save & Next

movavi Screen Recorder

A battery consists of a variable number (n) of identical cells, each having an internal resistance r connected in series. The terminal of the battery is short-circuited. A graph of current versus the number of cells will be as shown in figure.

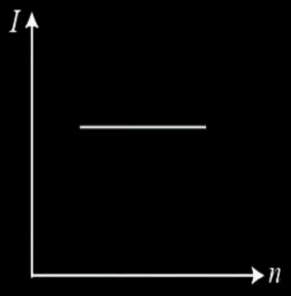


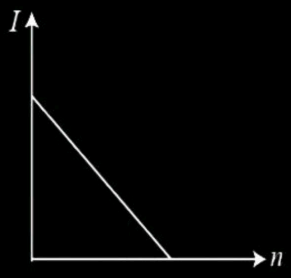


Mark for Review & Next | Clear Response

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Mark for Review & Next | Clear Response

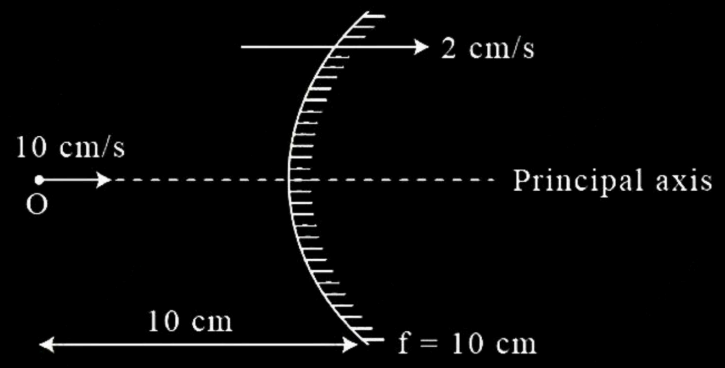
movavi Screen Recorder

Physics Numerical (Maximum Marks: 20)

Numerical Type

This section contains 5 questions. 5 are numerical questions. The answer has to be filled into the input box provided below.

What is the velocity (in cm/s) of image in situtation shown below. (O-object f = focal length). Object moves with velocity 10 cm/s and mirror moves with velocity 2 cm/s as shown.



Backspace

Mark for Review & Next

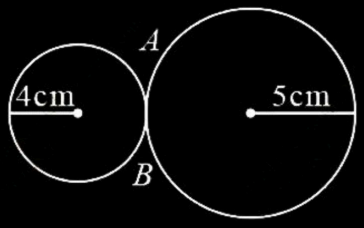
Clear Response

movavi Screen Recorder

Physics Numerical (Maximum Marks: 20)
Numerical Type

This section contains 5 questions. 5 are numerical questions. The answer has to be filled into the input box provided below.

Two soap bubbles of radii r_1 and r_2 equal to 4 cm and 5 cm respectively are touching each other over a common surface AB (shown in figure). Its radius will be (in cm)



Backspace

7 8 9
4 5 6
1 2 3

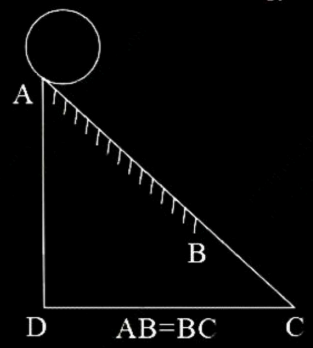
Mark for Review & Next | Clear Response | Save & Next

movavi Screen Recorder

Physics Numerical (Maximum Marks: 20)
Numerical Type

This section contains 5 questions. 5 are numerical questions. The answer has to be filled into the input box provided below.

Portion AB of the wedge shown in figure is rough and BC is smooth. A solid cylinder rolls without slipping from A to B. Find the ratio of translational kinetic energy to rotational kinetic energy, when the cylinder reaches point C .



Backspace

7 8 9

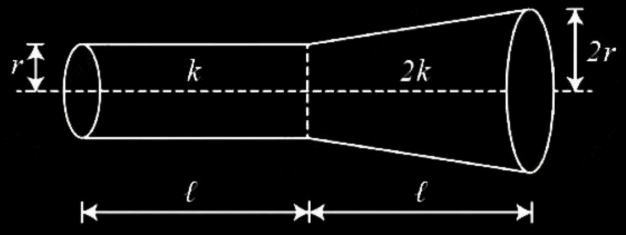
Mark for Review & Next | Clear Response

00:04:36
movavi Screen Recorder

Physics Numerical (Maximum Marks: 20)
Numerical Type

This section contains 5 questions. 5 are numerical questions. The answer has to be filled into the input box provided below.

A composite object is formed by combining a uniform rod of circular cross-section with thermal conductivity k and a frustum with thermal conductivity 2k as shown in the figure. The equivalent thermal conductivity of the object is given as $\frac{Nk}{5}$, find 'N'



Backspace

7 8 9

4 5 6

1 2 3

Mark for Review & Next

Clear Response

Save & Next

movavi Screen Recorder

Physics Numerical (Maximum Marks: 20)
Numerical Type

This section contains 5 questions. 5 are numerical questions. The answer has to be filled into the input box provided below.

A ball is projected upwards from the top of the tower with a velocity 50 m s^{-1} making an angle 30° with the horizontal. The height of tower is 70 m . After how many seconds the ball will strike the ground? (take value of $g = 10\ m/s^2$)

Backspace

7 8 9

4 5 6

1 2 3

0 . -

Left Right

Clear All

Mark for Review & Next

Clear Response

00:04:42

Save & Next

movavi Screen Recorder

Question No. 1 #6776c0da4a3f794b86e91b991

Section Details

Marking Scheme

Chemistry Single Correct (Maximum Marks: 80)
Only One Option Correct Type

This section contains 20 questions. 20 are multiple choice questions. Each question has multiple options out of which ONLY ONE is correct.

If the radius of electron orbit in a He^+ ion is 52.9 pm, the angular momentum of electron in that orbit is

- $\frac{h}{2\pi}$
- $\frac{h}{3\pi}$
- $\frac{h}{\pi}$
- $\frac{h}{4\pi}$

Mark for Review & Next

Clear Response

00:04:46

Save & Next

movavi Screen Recorder

Question No. 2 #6776c0da4a3f794b86e91b992

Section Details Marking Scheme

Chemistry Single Correct (Maximum Marks: 80)
Only One Option Correct Type

This section contains 20 questions. 20 are multiple choice questions. Each question has multiple options out of which ONLY ONE is correct.

Select the number of correct statements from the following:
(i) Delocalisation of σ electron is hyperconjugation
(ii) Delocalisation of π-electron is resonance
(iii) Inductive effect involves partial transfer of σ. electrons
(iv) Inductive effect involves partial displacement of σ elections
(v) Resonance effect is distance dependent
(vi) sp^3, sp^2 and sp hybridised carbon atoms can participate in resonance

- 3
- 4
- 2
- 5

Mark for Review & Next | Clear Response | Save & Next

movavi Screen Recorder

Question No. 3 #6776c0da4a3f794b86e91b993

Section Details | Marking Scheme

Chemistry Single Correct (Maximum Marks: 80)
Only One Option Correct Type

This section contains 20 questions. 20 are multiple choice questions. Each question has multiple options out of which ONLY ONE is correct.

Two separate experiments were carried out involving adiabatic reversible expansion, taking mono atomic gas in one case and a diatomic gas in the second case. In each case, the initial temperature was T_i and expansion was from volume V_1 to volume V_2. The final temperatures attained were T_M for mono atomic gas and T_D for diatomic gas. Then

- $T_M = T_D < T_i$
- $T_M < T_D < T_i$
- $T_D > T_M > T_i$
- $T_D < T_M < T_i$

Mark for Review & Next | Clear Response | Save & Next

movavi Screen Recorder

Question No. 4 #6776c0da4a3f794b86e91b994

Section Details Marking Scheme

Chemistry Single Correct (Maximum Marks: 80)
Only One Option Correct Type

This section contains 20 questions. 20 are multiple choice questions. Each question has multiple options out of which ONLY ONE is correct.

The screening effect of inner electrons will lead to

a. Decrease in the ionization potential.

b. Increase in the ionization potential.

c. Decrease in electron affinity.

d. Increase in electron affinity.

- ○ Both (a) and (c).
- ○ Both (a) and (d).
- ○ Both (b) and (c).
- ○ Both (b) and (d).

Mark for Review & Next | Clear Response

movavi Screen Recorder

Question No. 5 #6776c0da4a3f794b86e91b995

Section Details Marking Scheme

Chemistry Single Correct (Maximum Marks: 80)
Only One Option Correct Type

This section contains 20 questions. 20 are multiple choice questions. Each question has multiple options out of which ONLY ONE is correct.

Identify the correct statements from the following :

I. The ionic radius of Pr^{3+}, Dy^{3+} and Sm^{3+} follow the order, $Sm^{3+} > Pr^{3+} > Dy^{3+}$.

II. Eu^{2+} acts as strong reducing reagent.

III. Pu exhibits +7 oxidation state.

- I, II only
- I, III only
- I, II, III
- II, III only

Mark for Review & Next | Clear Response | 00:05:02 | Save & Next

movavi Screen Recorder

Chemistry Single Correct (Maximum Marks: 80)
Only One Option Correct Type

This section contains 20 questions. 20 are multiple choice questions. Each question has multiple options out of which ONLY ONE is correct.

Statement-1: If chlorine is passed into toluene at room temperature in the presence of Anhydrous $AlCl_3$, electrophilic substitution reaction takes place giving o- and p-chlorotoluenes.
Statement-2: If chlorine is passed through boiling toluene in the presence of UV light, substitution reaction in the aliphatic side chain occurs.

- Statement-1 is True; Statement-2 is True ; Statement-2 is a correct explanation for Statement-1.
- Statement-1 is True; Statement-2 is True; Statement-2 is not a correct explanation for Statement-1.
- Statement-1 is True; Statement-2 is False.
- Statement-1 is False; Statement-2 is True.

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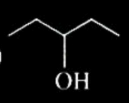
Question No. 7 #6776c0da4a3f794b86e91b997

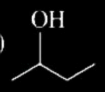
Section Details Marking Scheme

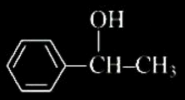
Chemistry Single Correct (Maximum Marks: 80)
Only One Option Correct Type

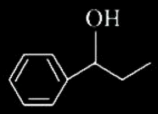
This section contains 20 questions. 20 are multiple choice questions. Each question has multiple options out of which ONLY ONE is correct.

Find out number of alcohols that can give positive iodoform test.

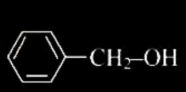
(a) 

(b) 

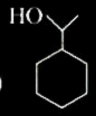
(c) 

(d) 

(e) $CH_3–CH_2–OH$

(f) 

(g) 

(h) 

- 3
- 4
- 5

Mark for Review & Next | Clear Response | Save & Next

movavi Screen Recorder

Question No. 8 #6776c0da4a3f794b86e91b998

Section Details Marking Scheme

Chemistry Single Correct (Maximum Marks: 80)
Only One Option Correct Type

This section contains 20 questions. 20 are multiple choice questions. Each question has multiple options out of which ONLY ONE is correct.

The correct sequence of decreasing number of π-bonds in the structures of H_2SO_3, H_2SO_4 and $H_2\ S_2O_7$ is :

- $H_2SO_3 > H_2SO_4 > H_2\ S_2O_7$
- $H_2SO_4 > H_2\ S_2O_7 > H_2SO_3$
- $H_2\ S_2O_7 > H_2SO_4 > H_2SO_3$
- $H_2\ S_2O_7 > H_2SO_3 > H_2SO_4$

Mark for Review & Next | Clear Response | Save & Next

movavi Screen Recorder

Question No. 9 #6776c0da4a3f794b86e91b999

Section Details Marking Scheme

Chemistry Single Correct (Maximum Marks: 80)
Only One Option Correct Type

This section contains 20 questions. 20 are multiple choice questions. Each question has multiple options out of which ONLY ONE is correct.

Which of the following reactions produce ethyl cyanide as a major product?

1. $CH_3CH_2CH = NOH \xrightarrow[K_2CO_3]{(CH_3O)_2O}$

2. $CH_3CH_2Cl \xrightarrow{\text{Ethanolic KCN}}$

3. $CH_3CH_2Cl \xrightarrow{\text{Ethanolic AgCN}}$

4. $CH_3CH_2CON(CH_3)_2 \xrightarrow[\text{Pyridine, } 70^\circ C]{C_6H_5SO_2Cl}$

- 1,3
- 2,3
- 3,4

Mark for Review & Next | Clear Response | Save & Next

movavi Screen Recorder

Question No. 10 #6776c0da4a3f794b86e91b9910

Section Details | Marking Scheme

Chemistry Single Correct (Maximum Marks: 80)
Only One Option Correct Type

This section contains 20 questions. 20 are multiple choice questions. Each question has multiple options out of which ONLY ONE is correct.

Pick the wrong statement from the following options:

- Deficiency of vitamin B6 (pyridoxime) results in convulsions
- Sources of vitamin are yeast, milk, green vegetables and cereals
- Deficiency of vitamin D cause Xerophthalmia
- Consumption of citrus fruits and green leafy vegetables in food prevents scurvy

Mark for Review & Next | Clear Response | Save & Next

movavi Screen Recorder

Question No. 11 #6776c0da4a3f794b86e91b9911

Section Details | Marking Scheme

Chemistry Single Correct (Maximum Marks: 80)
Only One Option Correct Type

This section contains 20 questions. 20 are multiple choice questions. Each question has multiple options out of which ONLY ONE is correct.

Column I	Column II
A. He	i. Highest electron gain enthalpy
B. Cl	ii. Most electropositive element
C. Cs	iii. Strongest reducing agent
D. Li	iv. Highest ionisation energy

The correct match of the contents in column I with those in column II is

- A-(iii), B-(i), C-(ii), D-(iv)
- A-(iv), B-(iii), C-(ii), D-(i)
- A-(i), B-(ii), C-(iii), D-(iv)
- A-(iv), B-(i), C-(ii), D-(iii)

Mark for Review & Next | Clear Response

movavi Screen Recorder

Question No. 12 #6776c0da4a3f794b86e91b9912

Section Details Marking Scheme

Chemistry Single Correct (Maximum Marks: 80)
Only One Option Correct Type

This section contains 20 questions. 20 are multiple choice questions. Each question has multiple options out of which ONLY ONE is correct.

For a chemical reaction $A \Rightarrow$ product, the mechanism of the reaction postulated was as follows:

$A(g) \underset{k_2}{\overset{k_1}{\rightleftharpoons}} 3B\ (g)\ ;\ B\ (g) \xrightarrow[RDS]{k_3} C\ (g)$

If the reaction occurred with individual rate constants k_1, k_2 and k_3, determine activation energy for the overall reaction if the activation energies associated with these rate constants are $E_{a_1} = 180\ kJ\ mol^{-1}$, $E_{a_2} = 90\ kJ\ mol^{-1}$ and $Ea_3 = 40\ kJ\ mol^{-1}$.

- $310\ kJ$
- $-10\ kJ$
- $130\ kJ$
- $70\ kJ$

Mark for Review & Next | Clear Response | Save & Next

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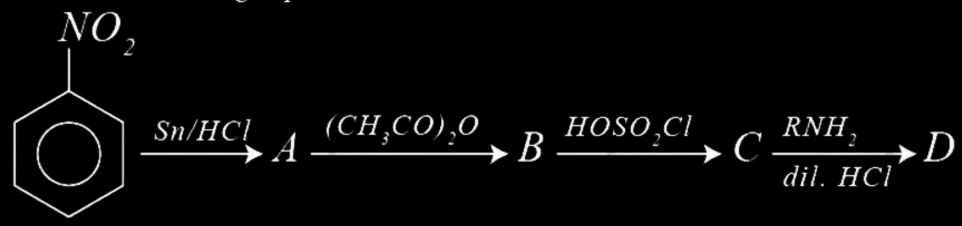
Question No. 13 #6776c0da4a3f794b86e91b9913

Section Details Marking Scheme

Chemistry Single Correct (Maximum Marks: 80)
Only One Option Correct Type

This section contains 20 questions. 20 are multiple choice questions. Each question has multiple options out of which ONLY ONE is correct.

Consider the following sequence of reactions:



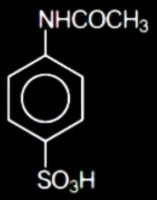
The correct structure of products (D) is

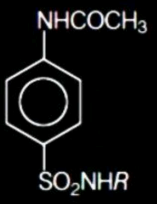


Mark for Review & Next | Clear Response | Save & Next

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Question No. 14 #6776c0da4a3f794b86e91b9914

Section Details | Marking Scheme

Chemistry Single Correct (Maximum Marks: 80)
Only One Option Correct Type

This section contains 20 questions. 20 are multiple choice questions. Each question has multiple options out of which ONLY ONE is correct.

Total number of optically inactive stereoisomers of succinic acid are

- 3
- 4
- 2
- 1

Mark for Review & Next | Clear Response | Save & Next

movavi Screen Recorder

Question No. 15 #6776c0da4a3f794b86e91b9915

Section Details | Marking Scheme

Chemistry Single Correct (Maximum Marks: 80)
Only One Option Correct Type

This section contains 20 questions. 20 are multiple choice questions. Each question has multiple options out of which ONLY ONE is correct.

In $2HI(g) \rightleftharpoons H_2(g) + I_2(g)$, 2 mole HI, 4 mole H_2 and 4 mole of I_2 are found at equilibrium. If some hydrogen is now introduced into the container such that at equilibrium concentration of HI and I_2 become equal. Find the Number of mole of H_2 added.

- 10 mole
- 20 mole
- 1 mole
- None of these

Mark for Review & Next | Clear Response | Save & Next

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Question No. 16 #6776c0da4a3f794b86e91b9916

Section Details Marking Scheme

Chemistry Single Correct (Maximum Marks: 80)
Only One Option Correct Type

This section contains 20 questions. 20 are multiple choice questions. Each question has multiple options out of which ONLY ONE is correct.

In the electrolysis of $CuSO_4$ solution, how many grams of Cu are plated out on the cathode in the time that it takes to liberate 5.6 litre of $O_2(g)$, measured at NTP at the anode?

- 31.75
- 14.2
- 4.32
- None of these

Mark for Review & Next | Clear Response | Save & Next

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Question No. 17 #6776c0da4a3f794b86e91b9917

Section Details Marking Scheme

Chemistry Single Correct (Maximum Marks: 80)
Only One Option Correct Type

This section contains 20 questions. 20 are multiple choice questions. Each question has multiple options out of which ONLY ONE is correct.

The IUPAC name of the following compound is



- 1-bromo-2-hydroxybenzonitrile
- 2-bromo-3-cyanophenol
- 2-bromo-3-hydroxy benzonitrile
- 3-hydroxy-2-brómobenzonitrile

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Chemistry Single Correct (Maximum Marks: 80)
Only One Option Correct Type

This section contains 20 questions. 20 are multiple choice questions. Each question has multiple options out of which ONLY ONE is correct.

For a aqueous metal ion d-d transition is observed by using visible light. Which is INCORRECT combination for absorbed light & observed colour.

Absorbed light	Observe colour
Yellow	Violet

Absorbed light	Observe colour
Orange	Blue

Absorbed light	Observe colour
Green	Red

Absorbed light	Observe colour
Blue	Green

Mark for Review & Next

Clear Response

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Question No. 19 #6776c0da4a3f794b86e91b9919

Section Details | Marking Scheme

Chemistry Single Correct (Maximum Marks: 80)
Only One Option Correct Type

This section contains 20 questions. 20 are multiple choice questions. Each question has multiple options out of which ONLY ONE is correct.

Arrange the following in increasing order of their boiling point.

$0.05\ \text{m}\ La(NO_3)_3,\ 0.05\ \text{m}\ MgBr_2,\ 0.05\ \text{m}\ NaNO_3$

- $0.05\ \text{m}\ NaNO_3 < 0.05\ \text{m}\ La(NO_3)_3 < 0.05\ \text{m}\ MgBr_2$
- $0.05\ \text{m}\ NaNO_3 < 0.05\ \text{m}\ MgBr_2 < 0.05\ \text{m}\ La(NO_3)_3$
- $0.05\ \text{m}\ MgBr_2 < 0.05\ \text{m}\ La(NO_3)_3 < 0.05\ \text{m}\ NaNO_3$
- $0.05\ \text{m}\ La(NO_3)_3 < 0.05\ \text{m}\ MgBr_2 < 0.05\ \text{m}\ NaNO_3$

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Question No. 20 #6776c0da4a3f794b86e91b9920

Section Details | Marking Scheme

Chemistry Single Correct (Maximum Marks: 80)
Only One Option Correct Type

This section contains 20 questions. 20 are multiple choice questions. Each question has multiple options out of which ONLY ONE is correct.

The hybridization states of the central atoms in the complexes $[Fe(CN)_6]^{3-}$, $[Fe(CN)_6]^{4}$ and $[Co(NO_2)_6]^{3-}$ are

- d^2sp^3, sp^3 and d^4s^2 respectively
- d^2sp^3, sp^3d and sp^3d^2 respectively
- d^2sp^3, sp^3d^2 and dsp^2 respectively
- all d^2sp^3

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Chemistry Numerical (Maximum Marks: 20)

Numerical Type

This section contains 5 questions. 5 are numerical questions. The answer has to be filled into the input box provided below.

Amongst the following, the total number of species NOT having eight electrons around central atom in its outer most shell, is

NH_3, $AlCl_3$, $BeCl_2$, CCl_4, PCl_5

Backspace

7 8 9

4 5 6

1 2 3

0 . -

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Clear All

Mark for Review & Next

Clear Response

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Chemistry Numerical (Maximum Marks: 20)
Numerical Type

This section contains 5 questions. 5 are numerical questions. The answer has to be filled into the input box provided below.

A and B decompose via first order kinetics with half-lives 54.0 min and 18.0 min respectively. Starting from an equimolar non-reactive mixture of A and B, the time taken for the concentration of A to become 16 times that of B in min. (Round off to the Nearest Integer).

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1 2 3

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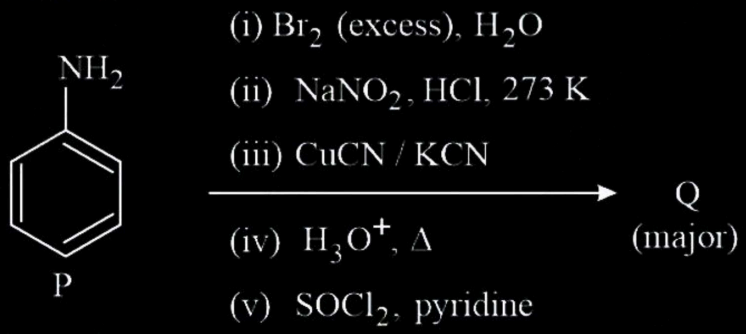
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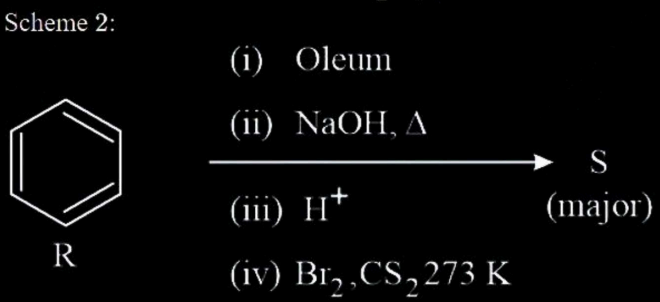
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Scheme 1 and 2 describe the conversion of P to Q and R to S, respectively. Scheme 3 describes the synthesis of T from Q and S. The total number of Br atoms in a molecule of T is

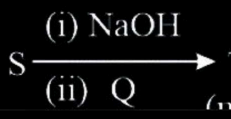
Scheme 1:



Scheme 2:



Scheme 3:



(major)

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R

(iii) H^+

(major)

(iv) Br_2, CS_2 273 K

Scheme 3:

$$S \xrightarrow[(ii)\ Q]{(i)\ NaOH} T \text{ (major)}$$

Backspace

7 8 9

4 5 6

1 2 3

0 . -

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Clear All

Mark for Review & Next

Clear Response

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Chemistry Numerical (Maximum Marks: 20)
Numerical Type

This section contains 5 questions. 5 are numerical questions. The answer has to be filled into the input box provided below.

$10mL$ of 0.2M solution of compound, $Na_2CO_3 \cdot NaHCO_3 \cdot 2H_2O$ is titrated against 0.05M HCl, y mL of HCl is used when phenolphthalein is used as an indicator and x mL of HCl is used when methyl orange is the indicator in two separate titrations. The value of $(x - y)$ will be

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4 5 6

1 2 3

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Clear All

Mark for Review & Next | Clear Response | Save & Next

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Question No. 5 #6776c0da4a3f794b86e91b995

Section Details Marking Scheme

Chemistry Numerical (Maximum Marks: 20)
Numerical Type

This section contains 5 questions. 5 are numerical questions. The answer has to be filled into the input box provided below.

0.20 g of an organic compound gave 0.12 g of AgBr By using Carius method, the percentage of bromine in the compound will be

Backspace

7 8 9

4 5 6

1 2 3

0 . -

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Clear All

Mark for Review & Next

Clear Response

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